

4.1

$$\begin{aligned}
 a. SST &= \sum (y_i - \bar{y})^2 \\
 &= \sum y_i^2 - N\bar{y}^2 \\
 &= 7825.94 - 20(19.21)^2 \\
 &= 445.458
 \end{aligned}$$

$$R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} \approx 0.8429$$

$$\begin{aligned}
 b. SSE &= SST(1 - R^2) \\
 &= 725.94(1 - 0.7911) \\
 &\approx 151.6489
 \end{aligned}$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{151.6489}{20-2} \approx 8.4249$$

$$c. R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} \approx 0.7105$$

4.3

$$a. \hat{y}_0 = 1.2 + 0.8(4) = 4.4$$

$$b. N = 5, \bar{x} = \frac{3+2+1+1}{5} = 1, \hat{\sigma}^2 = \frac{3.6}{5-1} = 1.2$$

$$\begin{aligned}
 se(f) &= \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} \\
 &= \sqrt{1.2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right]} \\
 &\approx 1.5875
 \end{aligned}$$

$$c. \alpha = 0.05 \text{ (two-tailed)}, t_{0.025;3} = 3.182$$

$$95\% PI = \hat{y}_0 \pm t_c se(f) = 4.4 \pm 3.182(1.5875)$$

$$95\% PI \approx (-0.651, 9.451)$$

$$d. \alpha = 0.005 \text{ (two-tailed)}, t_{0.005;3} = 5.841$$

$$99\% PI = 4.4 \pm 5.841(1.5875)$$

$$99\% PI \approx (-4.873, 13.673)$$

$$e. \hat{y} = 1.2 + 0.8(1) = 2$$

$$se(f) = \sqrt{1.2 \left[1 + \frac{1}{5} + 0 \right]} = 1.2$$

$$\alpha = 0.05 \text{ (two-tailed)}, t_{0.025;3} = 3.182$$

$$95\% PI = 2 \pm 3.182(1.2)$$

$$95\% PI \approx (-1.818, 5.818)$$

This interval is narrower than the interval computed in part (c), because the prediction is most precise when $x_0 = \bar{x}$.

a. log-linear model: $\ln(\widehat{PRICE}) = 4.3939 + 0.0360 SQFT$

Interpretation: We estimate a 1 unit increase in SQFT to increase expected price by 3.60%.

$$\overline{PRICE} = 250.2369, \quad \overline{SQFT} = 27.2821$$

$$\text{slope: } \frac{d(PRICE)}{d(SQFT)} = \beta_2 \cdot \overline{PRICE} = 9.020$$

$$\text{elasticity: } \beta_2 \cdot \overline{PRICE} \cdot \frac{\overline{SQFT}}{\overline{PRICE}} = \beta_2 \cdot \overline{SQFT} = 0.9834$$

At the sample means, we estimate that an added 100 square feet of interior space increases expected price by \$9020.

The elasticity at the sample mean is about 0.9834.

b. log-log model: $\ln(\widehat{PRICE}) = 2.0497 + 1.0248 \ln(SQFT)$

Interpretation: We estimate a 1% increase in SQFT to increase expected price by 1.02%.

$$\text{slope: } \frac{d(PRICE)}{d(SQFT)} = \alpha_2 \frac{\overline{PRICE}}{\overline{SQFT}} = 9.400$$

$$\text{elasticity: } \alpha_2 = 1.0248$$

At the sample means, we estimate that an added 100 square feet of interior space increases expected price by \$9400.

The elasticity at the sample mean is about 1.0248.

c. linear model: $\widehat{PRICE} = -115.4236 + 13.4029 SQFT$

$$R^2_{\text{linear}} = 0.6413$$

$$\sigma^2 = \frac{SSE}{N-2}, \quad \hat{y} = e^{\ln(\hat{y})}, \quad R^2_g = [\text{corr}(\hat{y}, y)]^2$$

$$R^2_g(\text{log-linear}) = 0.6622$$

$$R^2_g(\text{log-log}) = 0.6445$$

The generalized R^2 for the log-linear model is largest, so it fits the data better than the other two models.

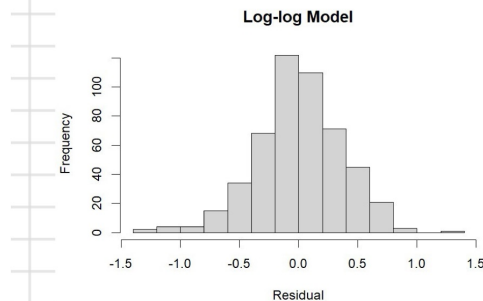
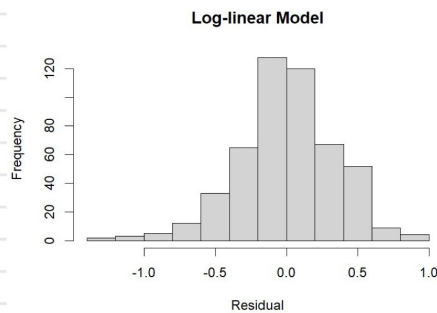
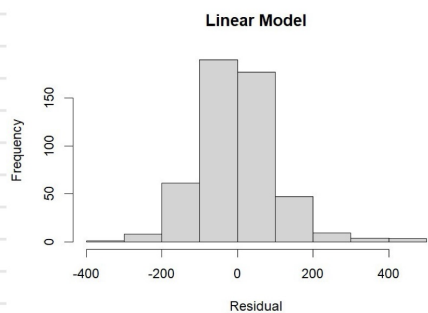
d. linear model: $JB = 221.1115$, $p\text{-value} < 0.0001$

log-linear model: $JB = 26.6790$, $p\text{-value} < 0.0001$

log-log model: $JB = 14.2787$, $p\text{-value} = 0.00079$

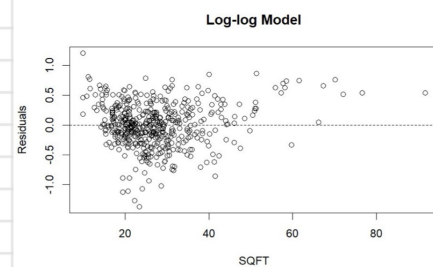
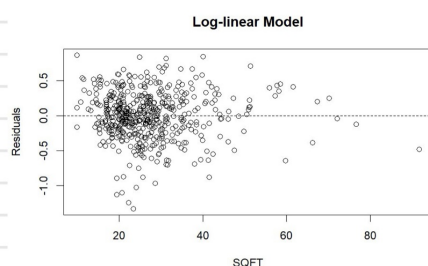
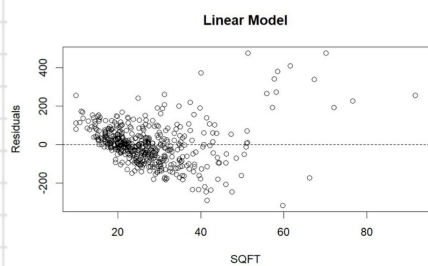
⇒ We reject normality of the regression errors at any level of significance.

```
> jarque.bera.test(resid(mod_linear))
Jarque Bera Test
data: resid(mod_linear)
X-squared = 221.11152, df = 2, p-value < 2.204e-16
> jarque.bera.test(resid(mod_loglinear))
Jarque Bera Test
data: resid(mod_loglinear)
X-squared = 26.678987, df = 2, p-value = 1.609651e-06
> jarque.bera.test(resid(mod_loglog))
Jarque Bera Test
data: resid(mod_loglog)
X-squared = 14.278711, df = 2, p-value = 0.0007932632
```



The residuals are not perfectly symmetric and display some skewness and tail behavior.

e.



It is fair to say that the logarithmic models provide less evidence of heteroskedasticity, as the residuals are more evenly spread across different values of SQFT.

f. linear model : \$ 246456

log-linear model : natural predictor : \$ 214234

corrected predictor : \$ 226982

log-log model : natural predictor : \$ 227539

corrected predictor : \$ 243151

```
> predict(mod_linear, newdata = newd)
1
246.4556636
> exp(predict(mod_loglinear, newdata = newd)) # natural
1
214.2336015
> exp(predict(mod_loglinear, newdata = newd) + sigma2_loglinear/2) # corrected
1
226.9817177
> exp(predict(mod_loglog, newdata = newd)) # natural
1
227.5386166
> exp(predict(mod_loglog, newdata = newd) + sigma2_loglog/2) # corrected
1
243.1514165
```

g. 95% prediction interval:

linear model : (\$ 44277 , \$ 448634)

log-linear model : (\$ 109768 , \$ 418117)

log-log model : (\$ 11141 , \$ 465841)

```
> predict(mod_linear, newdata = newd, interval = "prediction", level = 0.95)
1 fit lwr upr
1 246.4556636 44.27726567 448.6340616
> exp(predict(mod_loglinear, newdata = newd, interval = "prediction", level = 0.95))
1 fit lwr upr
1 214.2336015 109.7684718 418.1167439
> exp(predict(mod_loglog, newdata = newd, interval = "prediction", level = 0.95))
1 fit lwr upr
1 227.5386166 111.1406474 465.8405654
```

h. This is little difference between the log-linear model and log-log model. The log-linear model has a slightly higher generalized R^2 , but the log-log model has a slightly smaller Jarque-Bera statistic. The slopes and elasticities are similar. Overall, both models are preferable to the linear model.